Stratigraphy and paleomagnetism of the Jaw Face section, Wellsch Valley site, Saskatchewan[1]

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The basal part of Jaw Face section at the Wellsch Valley site, some 50 km north-northwest of Swift Current, Saskatchewan, contains mammalian fossils that are considered to be late Blancan to early Irvingtonian in age. It has been sampled for paleomagnetic studies through a thickness of 11 m. The section above 8.8 m has normal polarity; that below has reversed polarity. A tephra, which has yielded a minimum fission-track age of 0.69 ± 0.11 Ma, lies within the reversely magnetized part, near the top of the fossiliferous zone and just below the reversal at the 8.8 m level. Therefore, the change from normal to reversed polarity at the 8.8 m level is probably the Brunhes–Matuyama reversal at 0.73 Ma. Earlier studies have indicated that the Jaramillo (0.9–0.97 Ma) and Olduvai (1.67–1.87 Ma) Normal Polarity subzones occur within the lower predominantly reversed part of the Jaw Face section. In this more detailed study, we find no evidence of these normal subzones. We suggest that earlier results could have been artifacts of the procedures used. Several stratigraphic interpretations are possible. Our preferred interpretation is that the Jaw Face section extends from the upper part of the Matuyama Zone into the lower Brunhes Zone, that is, the mammalian faunas are entirely Irvingtonian in age. Alternatively, the section could contain a substantial hiatus, so the lower part would be early Matuyama in age and its faunas late Blancan – early Irvingtonisn, and the upper part could span the latest Matuyama and lower Brunhes zones. Other possibilities are discussed in the text.

La partie basale de la coupe de Jaw Face, exposée dans la vallée de Wellsch, à environ 50 km au nord-nord-ouest de Swift Current, en Saskatchewan, renferme des fossiles mammifères d'âge possiblement compris entre le Blancan tardif et l'Irvingtonien précoce. Elle a été échantillonnée pour une étude palémagnétique sur une épaisseur de 11 m. La coupe au-dessus le niveau de 8,8 m montre une polarité normale, mais en-dessous la polarité est inversée. Un tephra, qui a fourni un âge minimum déterminé à partir de traces de fission de 0,69 ± 0,11 Ma, est intercalé dans la partie caractérisée par l'aimantation inversée, près du sommet de la zone fossilifère et juste sous l'inversion au niveau de 8,8 m. Par conséquent, le passage de la polarité normale à inversée au niveau de 8,8 m correspond vraisemblablement à l'inversion Brunhes–Matuyama datée de 0,73 Ma. Les études antérieures ont indiqué que les sous-zones de polarité normale, Jaramillo (0,9–0,97 Ma) et Olduvai (1,67–1,87 Ma), se trouvaient dans la partie inférieure de la coupe de Jaw Face où domine la polarité inversée. Dans la présente étude plus détaillée, ces sous-zones de polarité normale n'ont pas été observées. Nous croyons que les résultats des études antérieures étaient des artefacts des méthodes utilisées. Plusieurs interprétations stratigraphiques peuvent être envisagées. L'interprétation qui nous semble la plus plausible décrit la coupe de Jaw Face comme une extension de la partie supérieure de la Zone de Matuyama dans la partie inférieure de la Zone de Brunhes, ainsi, les faunes mammifères seraient entièrement d'âge irvingtonien. D'autre part, cette coupe pourrait inclure une lacune de durée substantielle, et alors, l'âge de la partie inférieure de la coupe correspondrait au Matuyama, et les faunes mammifères appartiendraient au Blancan tardif – Irvingtonien précoce, la partie supérieure pourrait chevaucher la limite des zones de la fin du Matuyama et du Brunhes inférieur. D'autres interprétations sont discutées dans l'article.

[Traduit par la rédaction]

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Introduction

This work was undertaken as part of a programme to observe reversals of remanent magnetization in Early Pleistocene sequences in western Canada as an aid to their correlation. There are two important requirements. The first is the need to locate the last reversal between Matuyama and Brunhes polarity zones to serve as a time marker. The second is the need to determine if older normal magnetizations are present, magnetizations that could be correlated with the Olduvai or Jaramillo subzones within the reversed Matuyama zones, or with the underlying

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normal Gauss zone; if these could be established, then it would indicate the presence of very Early Pleistocene or Pliocene deposits. As we shall see the Matuyama–Brunhes boundary can be located in the Wellsch Valley site, but there is no clear evidence that older normal magnetizations are present.

The Wellsch Valley site (Fig. 1), near Swift Current, Saskatchewan, consists mainly of a sequence of unconsolidated silts and clays. The sediments forming the lower part of the sequence (unit I, Fig. 2) are considered to be sheetwash deposits laid down in a badland environment, whereas the remainder (units II–X) is composed of glacial and nonglacial deposits laid down in lakes and streams (Stalker 1971). The beds have yielded Early Pleistocene mammalian fossils; in particular, there are rodent fauna that have been assigned either an early Irvingtonian or a late Blancan age (Evernden *et al.* 1964; Stalker and Churcher 1972; Repenning 1980). The relationships amongst the stratigraphy, mammalian stages, and the reversal time scale are set out in Fig. 3.

Towards the top of the sequence (Fig. 2) and above the mammalian fossil beds, there is a tephra that has yielded a minimum date of 0.69 ± 0.11 Ma by the fission-track method (Westgate *et al.* 1978; Westgate and Gorton 1981). Electron spin resonance analysis of mammalian teeth has given an age of 0.28 ± 0.035 Ma, which is much less than those estimated from faunal and fission-track studies, presumably due to the late introduction of uranium (Zymela *et al.* 1988).

Previous paleomagnetic work has yielded normal polarity in the upper beds and predominantly reversed polarity in the beds below, and these have been correlated with the Brunhes and Matuyama polarity zones (Cox *et al.* 1967), respectively (Foster and Stalker 1976). G. Kukla of Lamont–Doherty Observatory collected on oriented sample from the tephra in 1974 and found it to be reversely magnetized (G. Kukla, personal communication, 1974). Within the lower reversed section, two levels of normal polarity were described by Foster and Stalker (1976) and assigned to the Jaramillo (Doell and Dalrymple 1963) and Olduvai (Gromme and Hay 1963; Gromme 1967) subzones. Later work by Barendregt (1984) confirmed the predominantly reversed nature of the lower beds and, he argued, provided further evidence for the presence of the Jaramillo and Olduvai subzones. However, neither study was based on extensive demagnetization experiments, and few details were reported. Without a detailed assessment of demagnetization characteristics, these sediments cannot be used confidently as recorders of the paleofield because they have a complex history of freezing and thawing and wetting and drying which may have modified or obliterated the original magnetization. As will become clear later, accurate assessment of polarity variations can only be made after detailed sampling and analysis. Hence our main purposes in resampling the site were to determine accurately the position of the Brunhes–Matuyama boundary in relation to the tephra and to investigate critically the evidence for normal horizons previously reported within the lower predominantly reversed section. These matters are of importance for dating the section and for making inferences about the distribution and migration of early Quaternary mammals across the prairies. If the normal horizons reported previously could be verified and definitely assigned to normal polarity subchrons within the Matuyama, then an age in excess of one million years should be assigned to them. If, on the other hand, these normal horizons do not survive further scrutiny, then it would indicate that the mammalian faunas could be late or middle Matuyama age.

Wellsch Valley site

The Wellsch Valley site lies in sections 4 and 9 of township 20, range 14, west 3rd meridian (Fig. 1), and about 10 km northwest of the town of Stewart Valley. Wellsch Valley is a broad coulee about 5 km long and 1 km wide, incised some 90 m below the prairie surface (Fig. 1). Most of the valley is cut in Quaternary glacial and nonglacial deposits, but its lower part, including the present stream notch, is cut in the Upper Cretaceous Bearpaw Formation (Caldwell 1968). This formation, and the glacial and nonglacial deposits above it, have been subject to slumping manifested by scarps (gravity faults), ridges and knolls (horsts), and elongated depressions (grabens) (Christiansen 1983; Sauer 1983). The Jaw Face section, which was sampled for paleomagnetic analysis, is on the west flank of a knoll and was chosen because the material shows little deformation, it is in its original stratigraphic position, and it contained mammalian fossils.

The original Welsch Valley may have been incised by meltwater from a retreating glacier or, less likely, during a halt in the advance of a glacier. Most probably, however, it was cut by a stream flowing into the valley of the South Saskatchewan River, which had been diverted to this area during a previous glaciation. Wellsch Valley was then filled with a variety of deposits that are now being eroded by the modern stream. They include gravel, unconsolidated silt and clay, minor sand, and till with interstratified silt and clayey silt (Fig. 2).

Interest in the Wellsch Valley site was sparked by W. Langston, formerly with the National Museum of Canada, who visited the site in 1959 while searching for dinosaur fossils in the Cretaceous bedrock. He found mammal bones on the surface, which he provisionally identified as "*Equus*?, *Camelops*, and an interesting ulna fragment from a wolf-sized animal" (W. Langston, personal communication to Stalker, 1960). Subsequently, Stalker examined the site and, together with Churcher, described the stratigraphy and made collections and fossil bones. All bones came from beds lying between bedrock and the lowest till (unit VIII) and generally from beds closest to bedrock.

Jaw Face section

The Jaw Face section (legal subdivision 8 of sec. 4, tp. 20, rge. 14, w. 3rd mer.; lat. 50°39′55″N, long. 107°52′30″W, Fig. 1) was first described by Stalker and Churcher (1972). It was so named because a dentary of *Camelops* was recovered, in 1966, about 7.0 m above the base of the exposure.

The sequence can be divided into 10 units. The lowest 1 m of unit I consists mainly of reworked local bedrock; it includes scattered stone bands, composed mostly of angular pieces of ironstone and petrified wood derived from local bedrock and rounded pebbles of quartzite and sandstone probably transported from the Rocky Mountains. A puzzling feature of unit I as a whole is the absence of stones derived from the Precambrian Shield. They would be expected because shield stones indicative of a Laurentide glaciation have been found in apparently corresponding deposits at the nearby New Mountain Section (Fig. 1). The deposits of unit I appear to have been laid down by sheetwash under fairly arid, apparently badland conditions. At times wind-blown material was added.

Unit I grades upward into the silt, clayey silty, and fine sand of units II–VI. Evidently, drainage became blocked after deposition of unit I, and the slow accumulation of fine sediment in quiet ponds commenced. Ponding may have been caused by

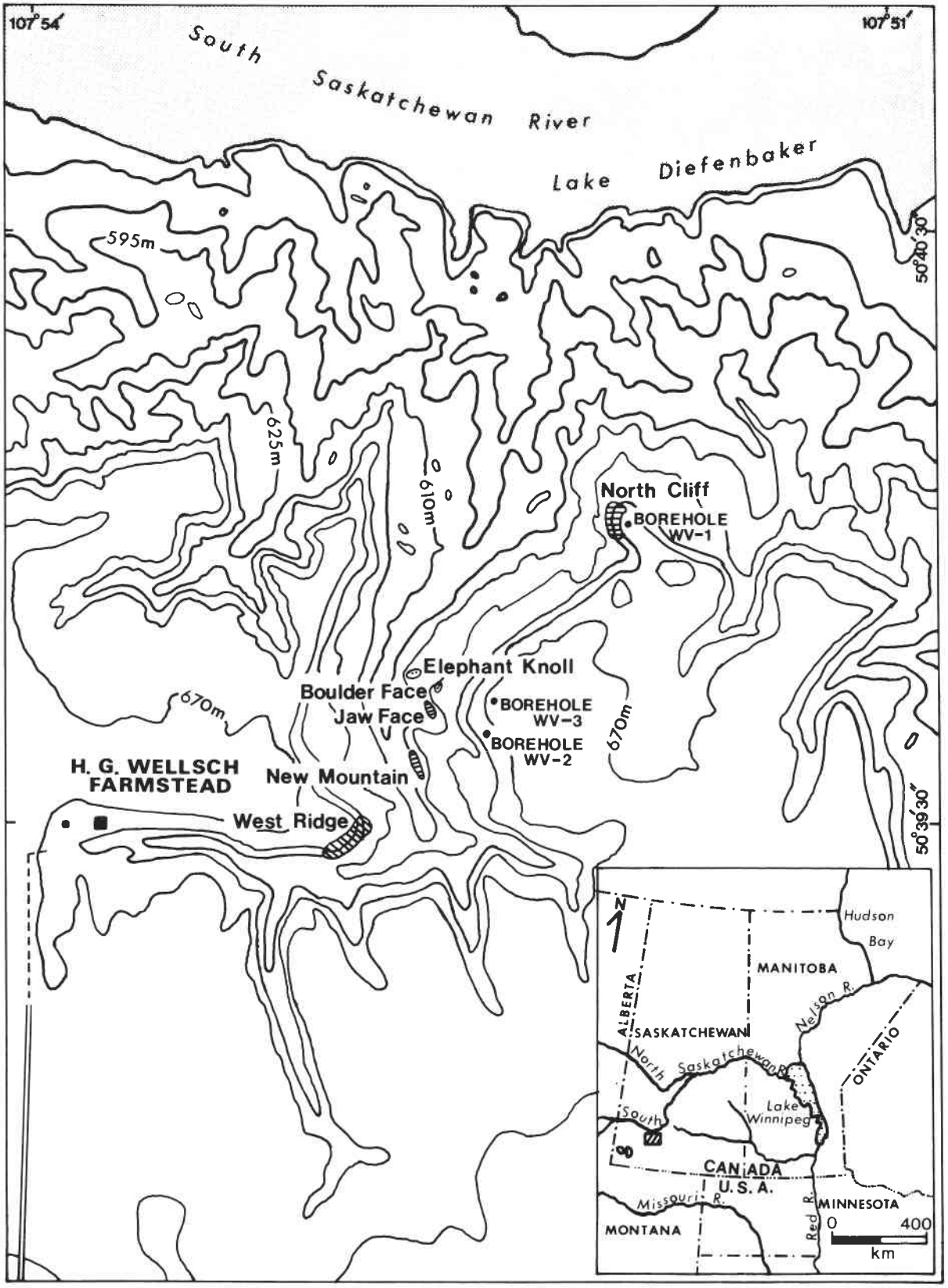


FIG. 1. Location map showing Jaw Face section at Wellsch Valley site.

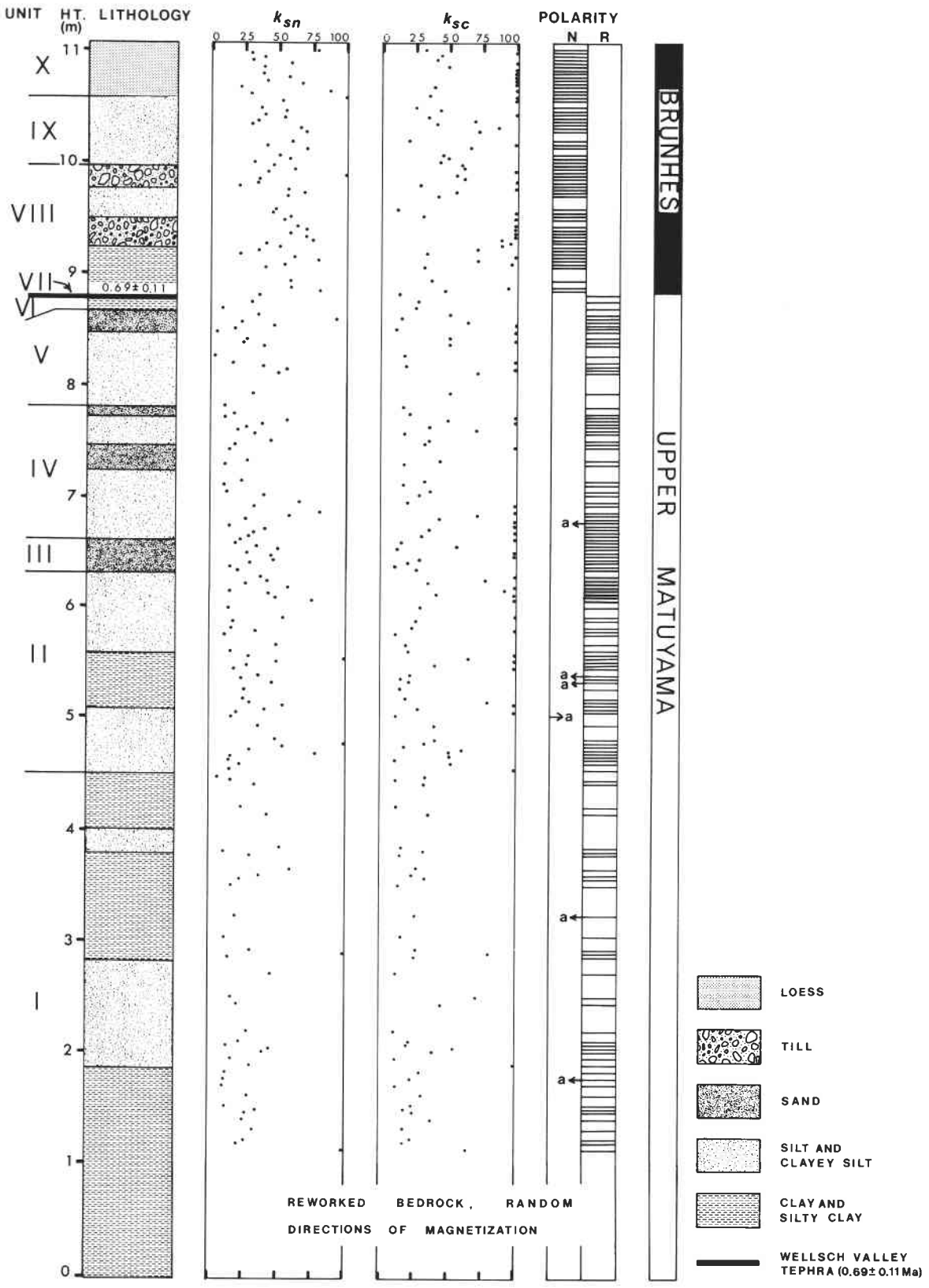


FIG. 2. Stratigraphy of Jaw Face section. The fission-track date from the tephra is after Westgate *et al.* (1978) and Westgate and Gorton (1981). Polarities determined from A- and B-type magnetizations are plotted. Option 1 (see Fig. 3) correlation with global time-scale is shown on the right. Anomalous indeterminate directions are labelled *a*. Fisher precision parameters k_{sn} for natural remanent magnetization and k_{sc} for cleaned directions of each specimen are plotted on right. Height is in metres above bedrock contact.

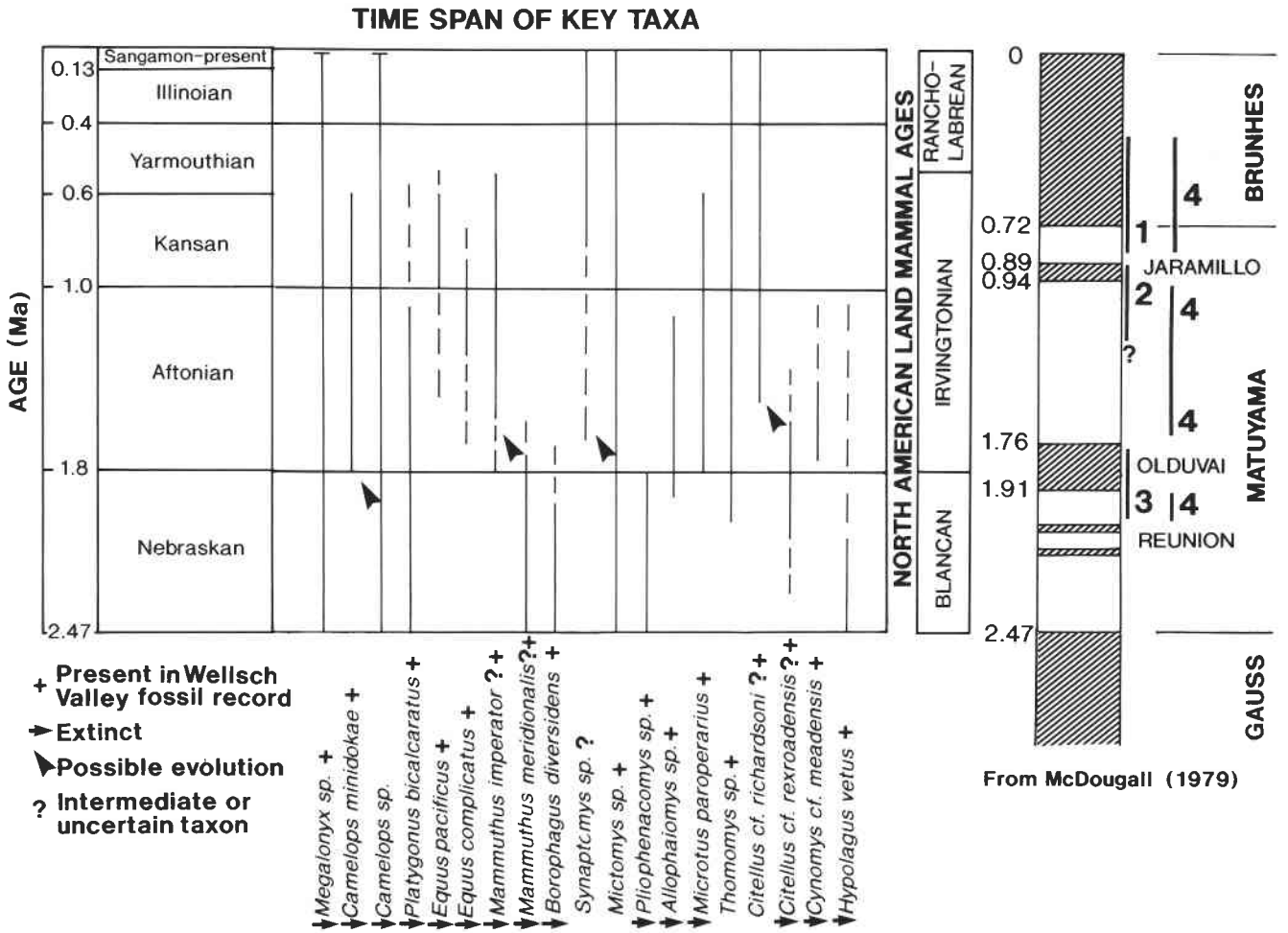


FIG. 3. Quaternary time-scales and ranges of taxa at Wellsch Valley, modified from Stalker and Churcher (1982). Polarity time-scale after Mankinen and Dalrymple (1979) and McDougall (1979); land-mammal ages after Repenning (1980); temporal ranges of mammalian taxa modified from Lundelius *et al.* (1987). Normal polarity is hachured, and reversed polarity white. Vertical bars 1–4 mark the four possible correlations of the Jaw Face section as discussed in the text. Our preferred options are 1 or 4.

blocking of the local stream valley by a glacier, although it is difficult to envisage a nearby glacier remaining sufficiently static to dam a lake for a long time without either overrunning the site or contributing stones and other debris. Lake level, however, did fall on several occasions (the lake may even have vanished at times), and fluvial sedimentation began. The evidence for this is sandy lag concentrates, presumably produced as a result of erosion of earlier deposits by small, sluggish, intermittent streams. Rodent bones and teeth, which originally had been diffusely scattered through the lake deposits, became concentrated in silts of unit II and sand beds of unit III. The hiatuses in deposition marked by these lag beds undoubtedly were brief, each ending when the lake again rose and quiet-water deposition recommenced. All told, there were probably three or four such cycles. Most larger bones came from unit IV, whereas the rodent bones and teeth came from units II and III (Fig. 2).

Unit VII is a 0.1 m thick tephra. It is thicker than other ash beds in southern Saskatchewan, and much of it is remarkably coarse, bearing in mind that the source is distant. Presumably much of it was washed from surrounding high ground and sorted and concentrated in this low spot. The lower part of the tephra is fairly pure, but the upper part is mixed with silt, commonly occurring in pockets, further supporting the idea that streams actively concentrated tephra at this site. Mud cracks occur on the surface of the underlying silt, indicating that the lake level was low at the time of the ash fall.

The tephra has been analyzed by J. A. Westgate (Stalker 1971, 1972; Stalker and Churcher 1972; and Westgate *et al.* 1978), who suggested that its composition indicates that it came from volcanoes of the Cascade Mountains. He notes that it does not resemble any other tephra so far described from the Canadian prairies.

Above the tephra, the silt in the lower part of unit VIII records a restoration of the lake and a return to quiet-water sedimentation. No vertebrate fossils have been found in this unit, perhaps reflecting the onset of frigid conditions that presumably impoverished the local faunas. The lake was short-lived, and coarsening of the material in the lower part of unit VIII heralded the approach of the glacier that shortly thereafter covered the area and deposited the tills. As this glacier receded, the proglacial lake reformed and lasted for a short time, until disrupted by renewed glaciation. The varved sequence of unit IX, seen at and near the surface in the Wellsch Valley area today, represents the last proglacial lake to cover the area. As this lake drained and conditions became drier, the loess deposits of unit X were laid down and the present Wellsch Valley drainage was established.

The sediments are generally buff or brown in colour due to oxides of iron. It is not clear from their appearance in the field to what extent these iron oxides were incorporated originally in the sediment in their present form, or were redistributed within the sediment during subsequent wetting and drying.

Paleomagnetic sampling and measurement

In July 1988, a section through Jaw Face was cleaned by backhoe, exposing 11 m of sediment (Fig. 2). A series of steps with vertical faces were carved with shovel and knife. Plastic cubes (2.4 cm sides; 357 altogether) were pressed into the vertical face in as continuous a fashion as possible.

Measurements were made on Schonstedt spinner magnetometers to which were attached devices that automate specimen manipulation and data acquisition. Batches of 15 specimens can be measured unattended. The system is controlled by IBM PC's. Stepwise alternating-field (AF) demagnetization was carried out using a Schonstedt GSC-5 for fields up to 100 mT and a Sapphire Instrument SI-4 for fields up to 180 mT. Demagnetization was carried out by treatment along three axes successively. With the exception of specimens showing very high internal dispersion, all have been treated in at least five sequential demagnetization steps. Specimen precision refers to the homogeneity of magnetization of specimens (Collinson *et al.* 1957) which has been estimated in the following way.

The magnetometers provide 12 readings for each specimen, from which four independent determinations of the magnetization vector are obtained. A measure of within-specimen homogeneity can be obtained by calculating the Fisher precision, k_{sn}, of the four determinations. For example, Quaternary basalts, which are generally homogeneous magnetically, yield k_{sn} values exceeding 500, whereas k_{sn} values for Jaw Face sediments rarely exceed 50 and are very commonly 20 or less. The value obtained for k_{sn} will depend on the dispersion of directions of magnetization of individual particles, on the number of particles in a specimen and their intensity, and on the sensitivity of the magnetometer used. As particle intensity diminishes and as instrumental noise increases, k_{sn} will fall. Although the separation of the various factors affecting k_{sn} is an interesting problem, its solution is not directly relevant because in the present context we simply use k_{sn} as an estimate of the precision of measurements whatever the sources of error might be. In Jaw Face sediments, there is a general decrease of k_{sn} with depth (Fig. 2). As will be shown, specimens with relatively high k_{sn} values generally have single-component magnetizations, which can be considered as reliable recorders of polarity. At the other extreme, magnetically incoherent specimens with k_{sn} of 10 or less provide no information about the paleofield. In Fig. 2, k_{sc} is the value obtained after magnetic cleaning. Note that k_{sc} is often higher than k_{sn}, indicating that cleaning often improves the specimen precision.

Magnetization types

In our experience, variable and often low paleomagnetic reliability is a common characteristic of early Quaternary sediment. Typical collections contain many specimens that are either poor recorders of the paleofield or do not record it at all. The main technical task when studying such material is to categorize specimens according to their reliability as field recorders. Four categories (A, B, C, D) were found to be sufficient to describe adequately the range of properties.

We began by measuring the natural remanent magnetization (NRM) of all 357 specimens and tabulating the k_{sn} values. We

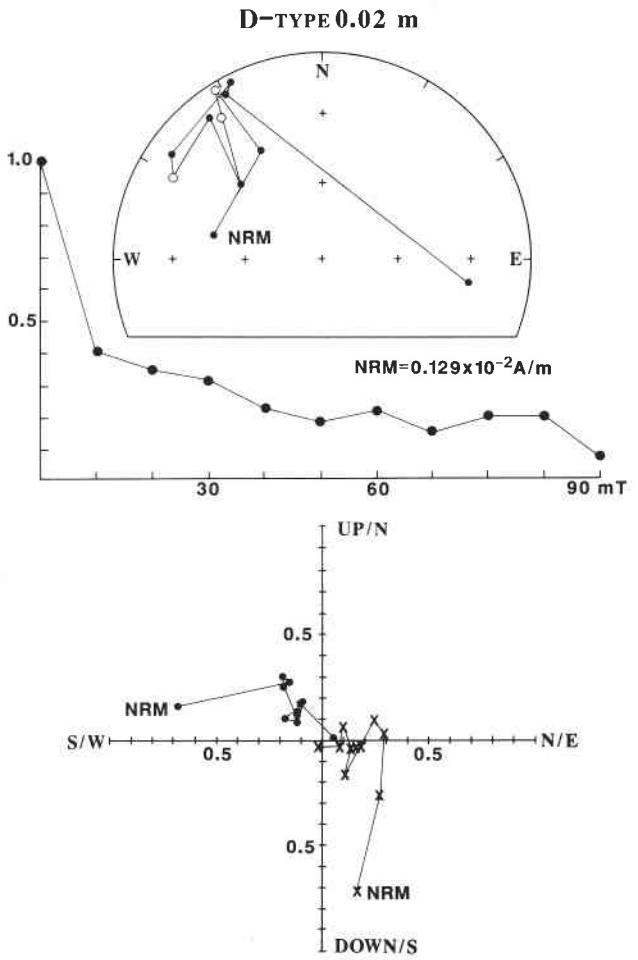


FIG. 4. D-type magnetization from reworked bedrock at 0.02 m. Directional changes (top) and normalized intensities (middle) during alternating-field demagnetization are shown. At the bottom are orthogonal plots on the north–south vertical plane (●) and the horizontal plane (+). Closed (open) symbols in the stereogram indicate downward (upward) inclination. Data are uninterpretable.

then selected 40 specimens for detailed study, making 10 demagnetization steps for each. It was found that the initial specimen precision, k_{sn}, provided a good indication of the accuracy with which magnetizations could be obtained. It was found, for example, that most specimens with k_{sn} of 10 or less did not yield accurately determined magnetizations. An example is given in Fig. 4. showing the scattering of directions during demagnetization and the absence of decay lines on orthogonal plots. Hence all specimens with $k_{sn} < 11$ were designated type D and were not studied further. There were 71 such specimens, making up 22% of the collection.

In other pilot specimens a great variety of demagnetization characteristics were encountered, and a search was instituted, using end-point determinations and line fitting of orthogonal plots, to establish a simple method that would provide reliable estimates of polarity. It was found that end points, although sometimes very rough end points, could be achieved by demagnetization in a range of alternating fields from 10 to 100 mT. A further batch of specimens (118) spread through the collection were then selected and treated at several steps (usually five) in this range, and it was established that approximate end points were generally present. All records for the 118 specimens and those from the pilot specimens were inspected, and it was found

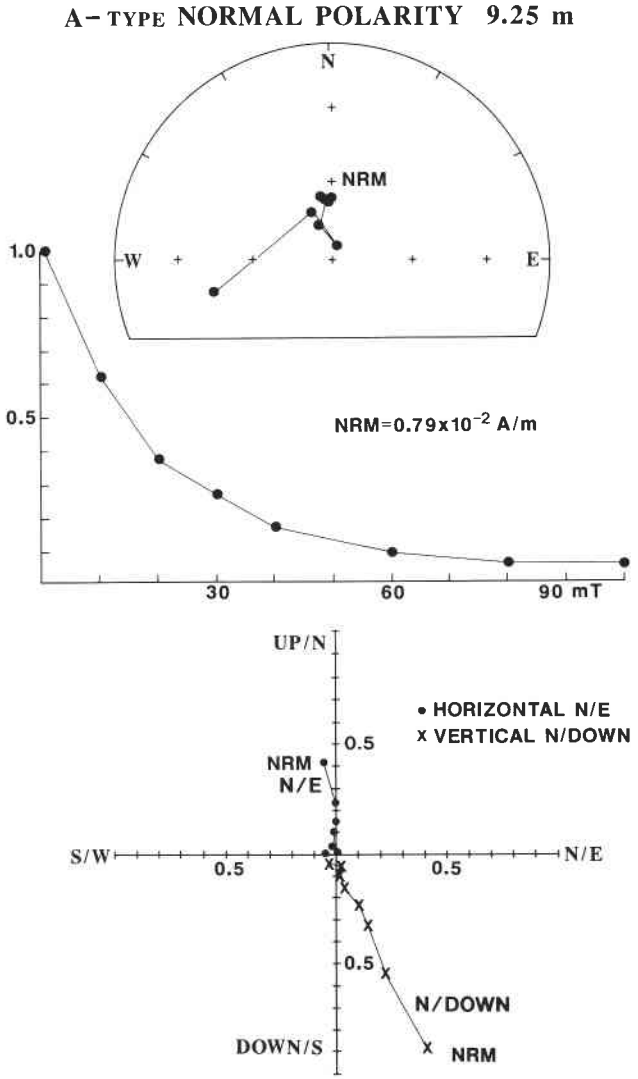


FIG. 5. Normal polarity, A-type magnetization, from oxidized silty clay layer at 9.25 m. Symbols as in Fig. 4.

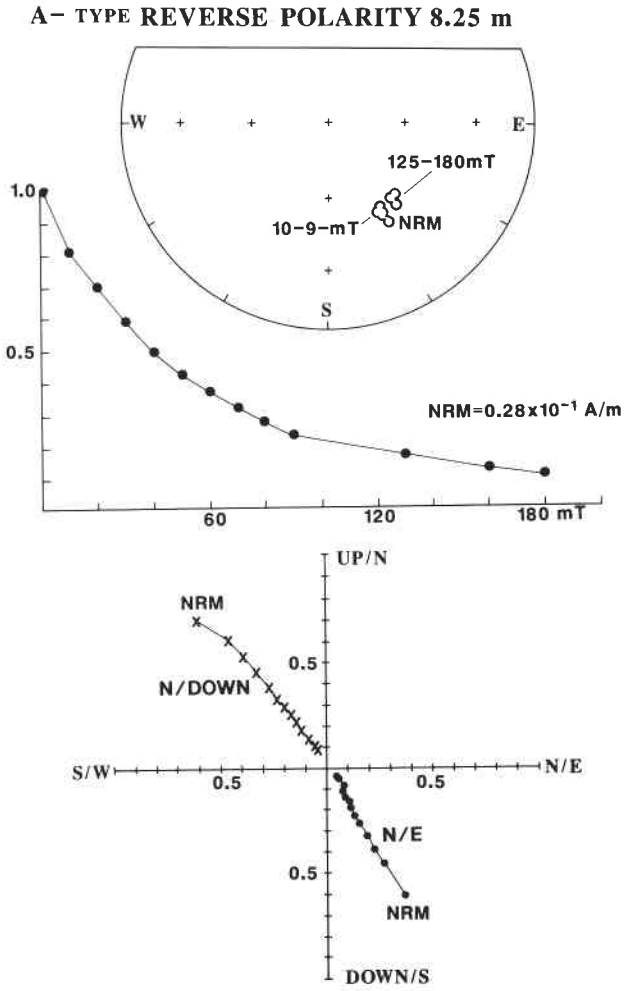


FIG. 6. Reversed polarity, A-type magnetization, from clayey silt layer at 8.25 m. Symbols as in Fig. 4, except inclinations are upward.

that polarity of these could be determined reliably by demagnetization at two levels. These were 20 mT for the upper part of the sequence (which as we shall see later was normally magnetized) and 40 mT for the lower part (which was reversed). The remainder of the collection (128 specimens) was then treated in these fields. The base of the section is composed of reworked bedrock (Figs. 2, 10). The directions are random (n = 32; k = 2) and provide no polarity information.

The specimens (excluding D-type, and those from the reworked bedrock), 254 all told, were then divided into three categories. Specimens in which the precision exceeded 30 after cleaning were defined as type A. There are 130 such specimens (36% of collection). Examples of A-type magnetization with both normal and reversed polarity are given in Figs. 5 and 6. Demagnetization curves are smooth, orthogonal plots move linearly to the origin, and directions change little throughout the demagnetization range. Sometimes, direction migrations do occur and orthogonal plots have hooked forms; the first part of the orthogonal plot shows the removal of magnetizations directed approximately along the present field (Brunhes overprint), and the second part decays linearly towards the origin. Least-squares fits to these lines, generally in the range of 20–100 mT but sometimes extending up to 180 mT, provide estimates of the paleodirections, with an average accuracy of 4° (expressed as the radius of the circle of confidence P = 0.05).

In 66 specimens (18% of collection) precision never fell below 11 during demagnetization but after cleaning did not exceed 30. These are assigned to the B category. In B-type specimens, although the magnetization directions cannot be accurately defined, the polarity is clearly indicated. An example is given in Fig. 7. The decay lines on orthogonal plots, although not well defined, are recognizable, and the directions, although only roughly grouped, are persistent up to high alternating fields in excess of 100 mT.

The remaining 58 specimens (16%) are defined as type C. Their precisions, although exceeding 10 initially, decrease during demagnetization, and magnetization directions could be determined only very roughly. An example is shown in Fig. 8. There are no recognizable decay lines on orthogonal plots, and the directions are erratic. However, they are systematically reversed, so there is an indication of polarity, although it is not well defined.

The directions of magnetization are plotted in Figs. 9 and 10 by categories, normal and reversed, respectively. The normal A-type magnetizations show substantial scatter, and their mean is about 10° shallower than that expected for a geocentric axial dipole (Table 1). The B- and C-type normal magnetizations are few, and the latter are widely dispersed.

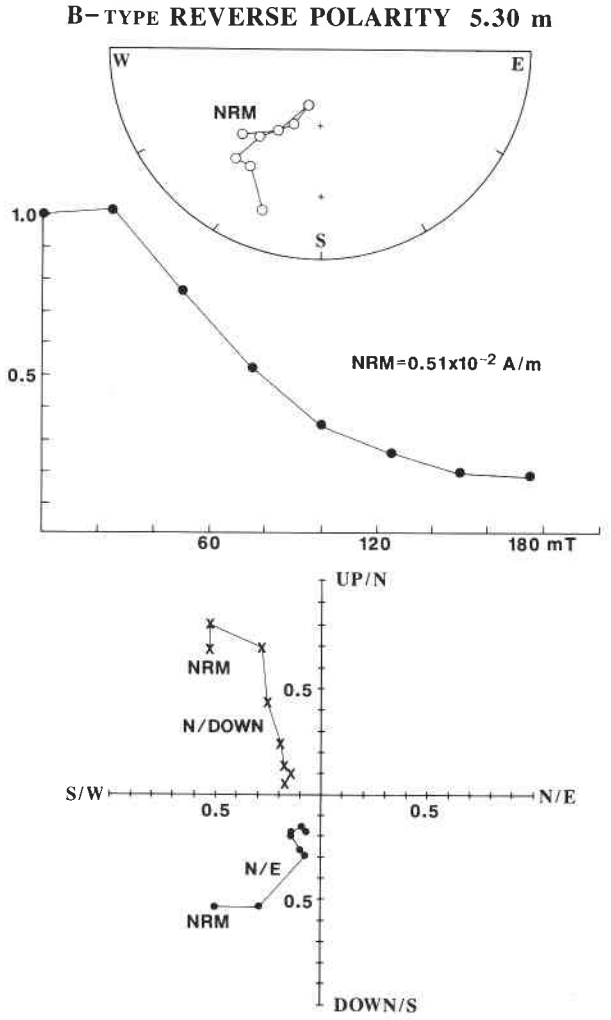


FIG. 7. Reversed polarity, B-type magnetization, from silt at 5.30 m. Symbols as in Fig. 4, except inclinations are upward.

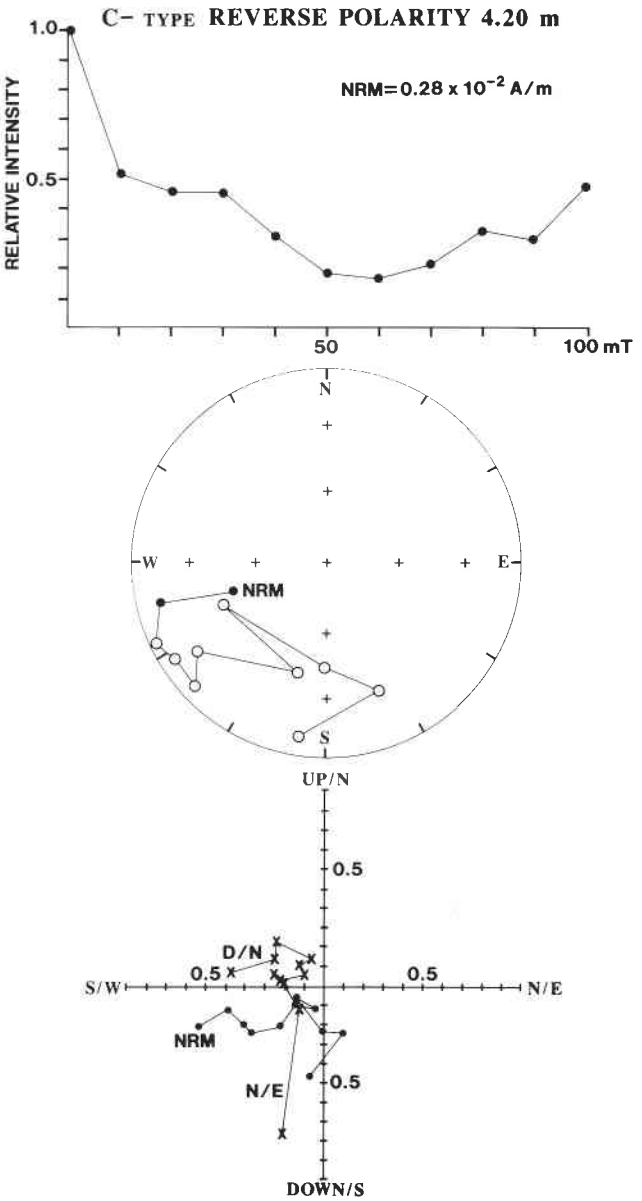


FIG. 8. C-type magnetization from silty clay at 4.20 cm. Symbols as in Fig. 4. Polarity is probably reversed.

The reversely magnetized A-type specimens also show substantial dispersion of directions. Dispersions are even greater in B- and C-types. Six specimens with B-type magnetization have directions that are neither normal nor reversed (Fig. 10). Their directions are 60° or more from the mean of a normal or reversed paleofield. These "anomalous" B-type specimens are distributed irregularly through the lower half of the section (Fig. 2) and hence do not indicate an excursion of the paleofield which would have yielded anomalous data concentrated at particular stratigraphic levels. They probably reflect incompletely resolved magnetizations. No anomalous magnetizations were found amongst A-type specimens. C-type magnetizations are more dispersed than B-type, and in many specimens the magnetizations present were probably not fully resolved. However, their mean does not differ significantly from that of the A- or B-type magnetizations (Table 1). To define polarity variations in the Jaw Face section only A- and B-types are used (Fig. 2). Normal magnetizations are confined to the section above the 8.8 m level, immediately above the ash (Fig. 2). Reversed polarity is confined to the lower part of the section. C-type magnetizations are predominantly from the lower part of the section, and although their polarity is less reliably determined, they are reversed. The mean inclinations of all types generally are lower than expected

TABLE 1. Summary of remanence directions

	M (A/m × 10^{-2})	D (°)	I (°)	n	k	α_{95} (°)	λ (°)	ϕ (°)
Normal								
A	0.76	9	58	56	18	5	77	40
B	0.60	341	65	4	17	23	77	153
C	0.55	244	78	4	2	75	37	226
Reversed								
A	0.44	199	−52	74	9	6	67	207
B	0.28	198	−49	56	7	8	65	212
C	0.22	194	−53	54	4	11	70	215

NOTES: A-, B-, and C-type magnetization as defined in text; M, intensity of magnetization; D and I, declination and inclination of the mean direction; n, number of specimens; k, precision parameter; α_{95}, circle of confidence ($P = 0.05$); λ and ϕ, latitude (north) and longitude (east) of the corresponding pole. The mean inclination expected for a geometric axial dipole is 67.7°.

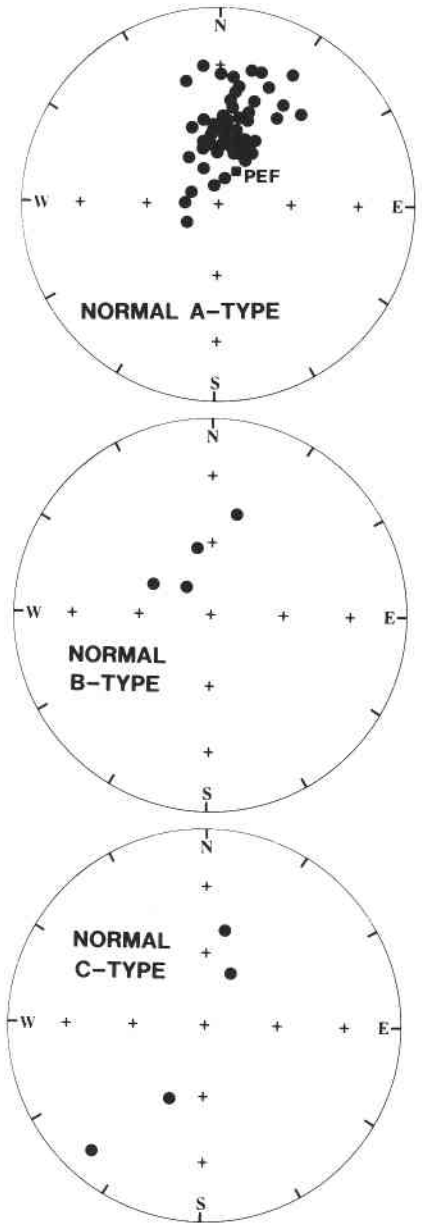


FIG. 9. Directions of normally magnetized specimens by type. Perimeter is present horizontal. Closed (open) symbols indicate downward (upward) inclination. PEF, present Earth's field.

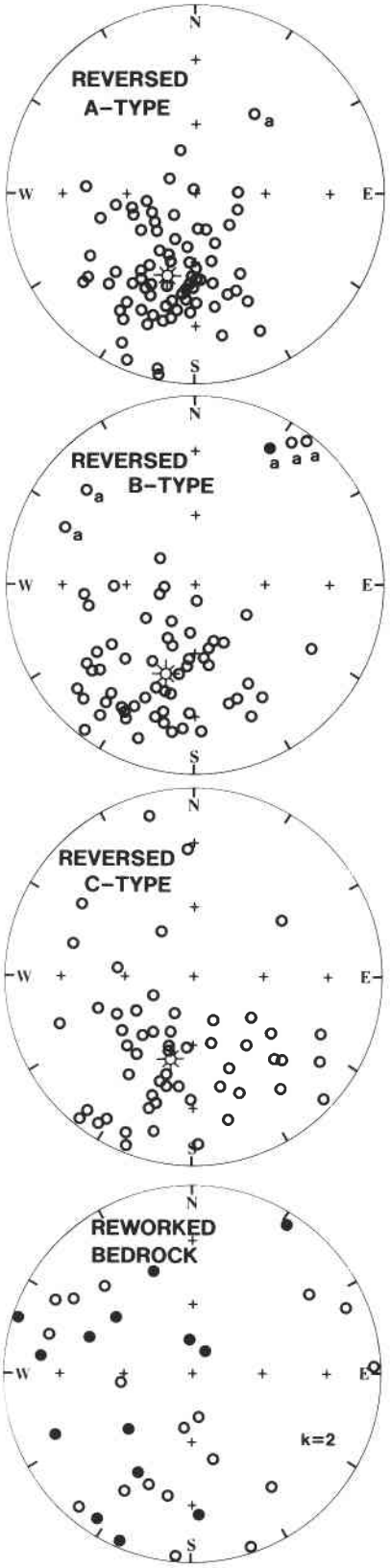


FIG. 10. Directions of reversely magnetized specimens by type. The bottom diagram gives directions from the reworked bedrock at the bottom of unit I. Anomalous directions are labelled *a*. See caption to Fig. 9.

for the geocentric axial dipole field. Departures for both polarities are significant (Table 1) and could have been caused by compaction upon dewatering of the sediments.

Reversed polarity directions are more dispersed than those of normal polarity. We suggest that the higher disperison of the reversed direction arises from contamination by small, unremoved Brunhes overprints. In many specimens these are remarkably tenacious and are not demagnetized in fields exceeding 100 mT, fields in which the underlying reversed magnetization is itself greatly diminished. If a small proportion of the overprint remains, then its effect would be much greater on the dispersion of reversed than normal magnetizations, a consequence of their near antiparallel and parallel alignments, respectively (Creer 1959).

The isothermal remanent magnetization (IRM) experiments

show that the sediments do not become saturated until 400 mT, indicating the presence of hematite, as would be expected from their buff and brown colour. There is therefore every likelihood that such sediments would acquire a high coercive force overprint, either as a viscous remanent magnetization (VRM) or as a chemical remanent magnetization (CRM) produced by the secondary redistribution of interstitial iron minerals during the frequent wetting and drying that the beds have undergone.

Discussion of the age of the Jaw Face sediments

The sedimentary units at the Jaw Face section cannot, at the moment, be directly traced to or correlated with beds of known age elsewhere. The nearest sections are along Swift Current Creek, about 12 km east of Wellsch Valley, where there is even less chronological control, and at Wascana Creek near Regina, Saskatchewan. At the latter locality, a volcanic ash, the Wascana Creek Ash of Christiansen (1961), which has a fission-track age of 0.6–0.7 Ma (Westgate *et al.* 1977), lies between two tills for which no regional correlation has yet been established.

The Wellsch Valley mammalian taxa identified at this locality provide strong clues to the age of the deposits (Stalker and Churcher 1982). The key taxa recovered from the site are listed in Fig. 3. The canid *Borophagus diversidens* is Blancan and usually Late Pliocene, after which time it is generally considered to have become extinct. The lagomorph *Hypolagus*, a Pliocene rabbit, is generally accepted as extinct by mid-Irvingtonian time. *Lynx*, the questionable Megalonychid sloth, the ground squirrel *Spermophilus*, the prairie dog *Cynomys*, the peccary *Platygonus*, and the camel *Camelops* are all first recorded in North America in Blancan deposits and, except for the sloth, peccary, and camel, persist to the present. The imperial mammoth *Mammuthus imperator haroldcooki* and the horses *Equus pacificus* and *Equus complicatus* were first recorded from the middle Irvingtonian, although the genus *Equus* (*Equus*) is itself first known about 3.3 Ma ago. The bog lemming *Synaptomys* (*Mictomys*) *kansasensis* was previously first reported from sediments dated at about 150 ka, although the genus is first recorded in North America about 2.5 Ma ago. The other cricetid rodents provide an even more precise indication of the age of the deposits, since new cricetid taxa seem to have appeared synchronously across the Holarctic landmasses. The voles *Microtus paroperarius* and *Allophaiomys ilanensis* first appear elsewhere in the Northern Hemisphere about 1.8 Ma ago, coincidental with the beginning of the Olduvai magnetic event (Repenning 1978). Thus the mammalian faunas at Jaw Face are in the Blancan–Irvingtonian age range. Evidence from the cricetid rodents, such as the vole *Pliophenacomys*, from *Borophagus*, and from some other taxa indicate that a late Blancan age is most likely. Such an age is permissible for all the taxa recorded, requiring temporal range extensions that are reasonable in the light of other known temporal ranges. However, the recorded age ranges of fossils are only estimates of their true range. All ranges are minimal and subject to extension in either direction as new specimens are discovered. Hence, although the most probable age of the Jaw Face faunas is late Blancan, this is not a definitive determination.

If this late Blancan age assignment is correct, then the Wellsch Valley site contains some of the earliest records in North America of true mammoths (*Mammuthus*), the bog-lemming *Synaptomys* (*Mictomys*), and the vole *Microtus*, as well as one of the last appearances of the bone-eating dog *B. diversidens* and the small rabbit *Hypolagus*.

Borophagus diversidens, Pliophenacomys sp., *Allophaiomys* sp., *Cynomys* cf. *meadensis*, and *Hypolagus vetus* are not reported from the North American fossil record (Fig. 2) after 1.0 Ma (middle Matuyama). Their presence at Jaw Face suggests that at least part of the reversely magnetized sequence there are deposited sometime between the Olduvai and Jaramillo normal polarity subchrons.

Amino-acid racemization studies have been carried out by J. L. Bada of Scripps Institute of Oceanography, La Jolla, California, on part of a humerus of a large horse, probably *E. pacificus*, from Wellsch Valley. The D/L aspartic acid ratio was 0.52, and the age obtained was 300 ka (J. L. Bada, personal communication to Stalker, 1979). However, the extent of racemization of other amino acids indicated that the bone contained secondary amino acids, and so 300 ka is only a minimum.

Paleomagnetic polarity provides further constraint on ages. The Jaw Face section consists of sediments that, at best, are only moderate indicators of the paleofield. This arises partly because of the common presence of a tenacious Brunhes overprint and partly because of the complex history of the older sediments (i.e., cryoturbation and repeated wetting and drying) which presumably has disturbed to varying degree magnetic orientations within the sediment. However, it does contain material that reliably records the polarity of the paleofield (A- and B-type data of Figs. 9 and 10), and there are sufficient data (196 *A*- and *B*-type specimens) to provide good coverage of the section (Fig. 2). The largest gaps in the record (10–20 cm) occur between the 2 and 4 m levels. Elsewhere, gaps are typically 5–10 cm. Some parts of the section, especially the 6–7 m level and above the 9 m level, are well covered with an average spacing of about 4 cm.

Up to a height of 8.8 m only reversed polarity is found. From there to the top of the section, the polarities are normal. The tephra (dated at 0.69 ± 0.11 Ma) could not itself be sampled at our section because it is too friable, but it has previously been reported to be reversely magnetized (see Introduction). Immediately above and below the tephra there are specimens with A-type magnetization and normal and reversed polarity, respectively, so the stratigraphic position of the reversal is well established. No transitional directions were recorded, indicating a gap in sedimentation at this level.

In earlier studies, normal polarities were described from specimens within the lower predominantly reversed part of the section (Foster and Stalker 1976; Barendregt 1984). In our study, the lower section contained no reliable evidence of normal magnetizations and the sequence of polarities given in Fig. 2 is much simpler than that described previously. We suggest that the positive (normal) inclinations previously recorded within the lower reversed section are not reliable indications of a normal polarity paleofield for three reasons. First, many examples of C- and D-type magnetizations actually come from levels comparable to those from which normal magnetizations have been reported. One such example is given in Fig. 4. If this specimen had been simply cleaned at 10 mT without demagnetization through a wider range (the procedure used by Barendregt 1984), then an apparent normal polarity would have been recorded. Such a reading would be spurious, because magnetization at higher fields reveals complex behaviour and neither direction nor polarity can be determined (Fig. 4). Second, to establish the existence of the Jaramillo and Olduvai subchrons it is necessary to show, by repetitive sampling, that true reversals of approximately 180° are present. Previous evidence from Jaw Face for the Olduvai is based on four samples giving virtual geomagnetic

poles (VGP's) about 60° from the present pole, and for the Jaramillo, three samples giving VGP's about 45° from the present pole. These departures from a true 180° reversal are too large in magnitude and the observations too few in number to constitute evidence for the presence of discrete polarity horizons. The observation of systematic and repetitive high-latitude (75° or more) VGP's is essential to demonstrate true reversal. These have not been observed. Third, in either studies at Jaw Face the Jaramillo is shown as a split event, whereas in the most recent global time scale, it is not (Fig. 3). Hence we conclude that previous records of normal polarity within the lower reversed section at Jaw Face could have been artifacts of the inadequate procedures used.

Conclusions

It is possible that the horizons immediately above the tephra belong to the lower part of the Brunhes Normal Polarity Zone and the beds beneath to the later part of the Matuyama Zone. This correlation with the global polarity time scale is shown as option 1 in Fig. 3. Other possibilities are that the reversed-to-normal transition observed at the 8.8 m level is either at the base of the Jaramillo (option 2) or the base of the Olduvai (option 3) subchrons. Finally, it is possible that the Jaw Face section contains a long depositional gap (disconformity) or gaps, so the upper units are located in both the upper Matuyama Reversed Polarity Zone and in the lower Brunhes Normal Polarity Zone, whereas the reversed horizons of lower units may belong to an earlier part, or parts, of the Matuyama Reversed Polarity Zone (option 4, Fig. 3). These four options indicate the inherent ambiguity in polarity studies. All options are compatible with the tephra fission-track age which is a minimum estimate only.

Of the four options discussed. The first provides the simplest explanations of the presently available paleomagnetic data. If option 1 is correct, then the reversal observed in the Jaw Face section is the Matuyama–Brunhes reversal and the mammalian faunas belong to the upper Matuyama Reversed Polarity Zone (the interval between the Jaramillo subzone and the Brunhes Zone). This would mean that the fossils are younger than the balance of the paleontological evidence, as presently interpreted, would indicate. If the fossils are truly late Blancan, then option 4 becomes more probable, and there is one or more disconformity within the Jaw Face section, the lower part being early Matuyama and the upper part late Matuyama to Brunhes in age. Although the paleomagnetic record found at the Jaw Face section substantiates the antiquity of the fossils found there and places them within the Reversed Matuyama Zone, it does not, however, determine where in this zone the Jaw Face fossil-bearing sediments lie.

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